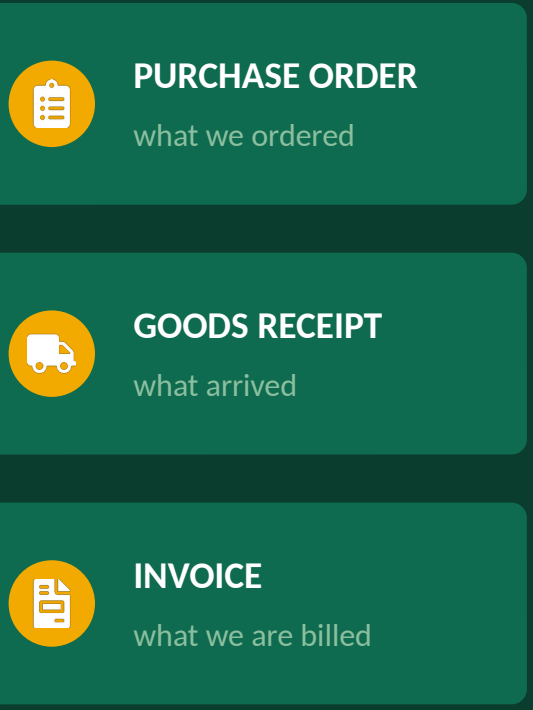
Agentic AI for Accounts Payable

Invoice three-way match with a team of cooperating agents

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Why invoices go wrong

Before paying a vendor, finance teams must check the invoice against the purchase order and the goods receipt. Done by hand, it is slow, and every miss becomes an overpayment.



Domain: Banking & Finance, Business Operations

What slips through

- **Price above the PO**
- **Billed more than received**
- **Same invoice paid twice**

From manual checks to agents

Today: manual review

- Staff compare three documents by hand
- Slow and repetitive for every invoice
- Errors often found after payment
- No consistent written reason

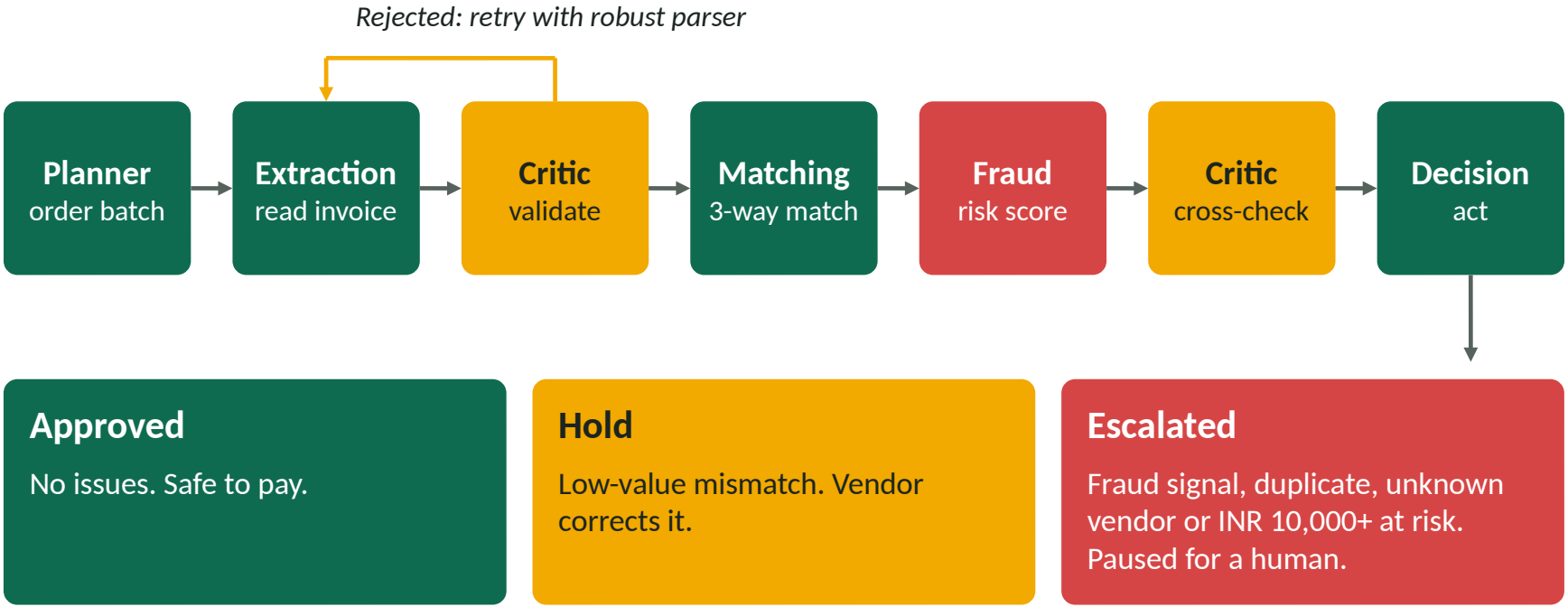
With agents

- Planner orders the batch by urgency
- Agents extract, validate and match
- Problems caught before payment
- Every decision comes with a reason

Seven agents, one job

Planner	Sequences the batch by due date and amount; flags urgent invoices
Extraction	Turns invoice text into a typed schema with an LLM or parser; retries if rejected
Critic	Rejects extractions that fail arithmetic checks; confirms every variance is explained
Matching	Compares invoice, PO and goods receipt on price, quantity, tax, vendor and duplicates
Fraud	Flags lookalike vendors, bank changes and amount spikes; scores risk
Decision	Applies business rules to approve, hold or escalate, and writes the reason
Human Review	Pauses escalations for a person, then updates vendor memory

How the agents hand off



Fraud Agent: what matching misses

+70

pay-to bank account differs from the one on file

+60

vendor name one letter off an approved vendor

+40

total above 3x the vendor's largest past invoice

+15

suspiciously round total

- Risk score 0-100: these points, plus up to 20 for a vendor whose invoices were often escalated. 40 or more escalates.
- Every point comes with its reason, so the reviewer sees exactly why an invoice was stopped.
- The spike rule was tuned on 200,000 simulated invoices: about 0.02% false alarms.

Human in the loop, with memory

1 Pause

LangGraph interrupt checkpoints the graph on escalation

2 Review

Approve, hold or reject in the Review queue

3 Resume

The graph continues from its checkpoint

4 Learn

The decision is saved to vendor memory

Real trace: a vendor's bank-account change, approved by a reviewer

```
Fraud Agent: BANK_CHANGE; risk score 70/100
Decision Agent: ESCALATED (fraud / control risk: BANK_CHANGE)
... graph paused at interrupt; invoice waits in the Review queue ...
Human Review: Saafalya reviewed: APPROVE -> APPROVED - "new account confirmed by phone"
              new bank account saved to vendor memory: next invoice is not flagged
```

System architecture

Streamlit UI
live agent trace, review queue, PDF upload, results, vendor memory

LangGraph state graph
shared state, retry edge, interrupt + checkpointer for human review

Agents

Planner | Extraction | Critic | Matching | Fraud | Decision | Human

Tools and data
PO DB, GRN DB, vendor master, duplicate registry, vendor memory, rules

LLM
Ollama local model or mock

Data flow

1. Batch or PDF enters
2. Graph shares state
3. Agents use tools and memory
4. Escalations pause for a human
5. Decision goes to the UI

Demo walkthrough

1 Generate batch

24 synthetic invoices, 8 seeded errors, 3 of them fraud

2 Run agents

Watch each agent step live

3 Review queue

Approve, hold or reject paused invoices

4 Upload PDF

Run the agents on a PDF invoice

Live trace for one invoice (real output)

```
Extraction Agent: attempt 1 via fast parser: INV-7009 | total 1.00
Critic Agent: extraction REJECTED (qty x unit != amount) -> retry
Extraction Agent: attempt 2 via robust parser: INV-7009 | total 1,734,135.13
Fraud Agent: AMOUNT_SPIKE (8.1x vendor's largest invoice); risk score 40/100
Decision Agent: ESCALATED (fraud risk score 40 >= 40) -> Review queue
```

What the validation shows

8/8

seeded errors caught

3/3

fraud cases escalated

0/16

clean invoices wrongly flagged

99.98%

decided correctly across 37,200 generated invoices

- Default batch: 24 invoices, 8 seeded errors (3 fraud) and 16 clean; INR 0.99L at risk flagged, equal to the seeded total.
- The Critic loop fixed misread amounts before matching. 15 automated tests pass.
- High scores reflect rule-based matching on known data, not real-world accuracy.

Limits and next steps

Limitations

- Synthetic data and one fixed invoice layout
- Rule-based matching scores optimistically
- No OCR for scanned PDFs, no ERP link or login
- Time savings are estimates, not measurements

Future scope

- OCR for scanned PDFs and photos
- ERP connectors such as SAP and Tally
- Benchmark LLM extraction against the parser
- Learn fraud weights from reviewer decisions
- Role-based approvals and MCP tools

Thank you

Questions?



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